

Molecular Dynamics: Simulations of Hydrated Biomolecules

PWF Summer school on Theoretical and Computational Biophysics

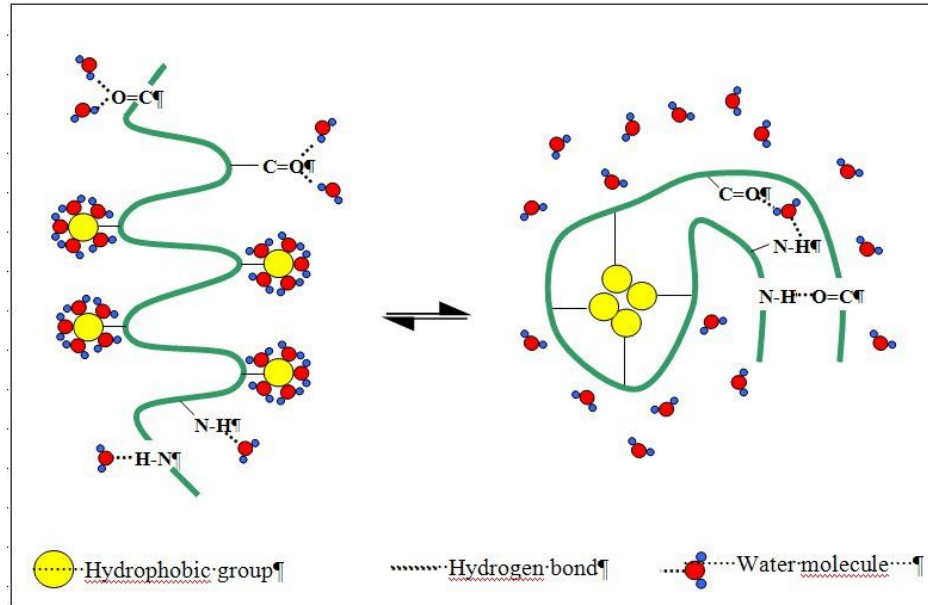
University of Colombo

Day 2/5

Dr. Marta Monti

Force Field

What about the parameters for H_2O ?



H₂O models

Models for H₂O can be distinguished by:

1. Number of sites (interaction points)
2. Rigidity
3. Inclusion of polarization effects

H₂O models

Models for H₂O can be distinguished by:

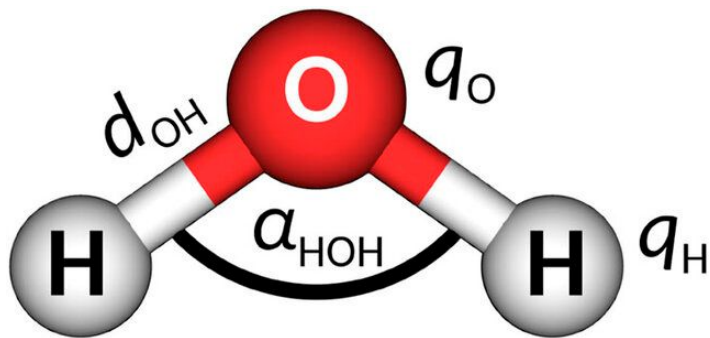
1. Number of sites (interaction points)
2. Rigidity
3. Inclusion of polarization effects

SPC and TIP3P

- Three-site approach: 3 interaction points corresponding to the 3 atoms of the molecule
- Each site has a point charge; LJ parameters for the O atom
- Rigid geometry fixed according to the model; TIP3P uses a geometry close to the experimental gas-phase structure, while SPC uses the ideal tetrahedral HOH angle.

H₂O models - TIP3P

TIP3P



```
[ moleculetype ]
; molname      nrexcl
SOL             2

[ atoms ]
; id   at type res nr  residu name   at name      cg nr  charge
1     opl_111  1      SOL           OW          1      -0.834
2     opl_112  1      SOL           HW1         1       0.417
3     opl_112  1      SOL           HW2         1       0.417
```

```
[ bonds ]
; i     j       funct  length  force.c.
1       2       1      0.09572  502416.0  0.09572      502416.0
1       3       1      0.09572  502416.0  0.09572      502416.0
```

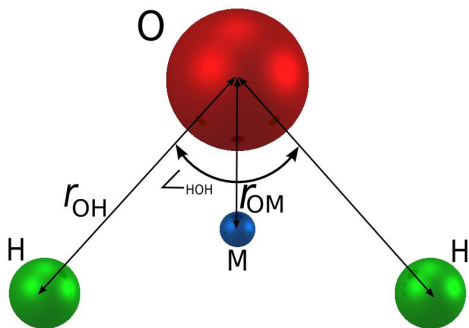
```
[ angles ]
; i     j     k       funct  angle  force.c.
2       1     3       1      104.52  628.02  104.52  628.02
```

Widely used and computationally efficient, but it overestimates water mobility

H₂O models

SPC/E, SPC/F, and TIP4P

- SPC/E model adds an average polarization correction and improves properties such as density and diffusion relative to SPC.
- SPC/F is a flexible variant that provides improved diffusion and dielectric properties compared with rigid SPC.
- TIP4P introduces a 4th interaction site carrying the negative charge along the HOH bisector, providing a more realistic electrostatic representation.



Different models reproduce different properties — there is no universally “best” water model.

Why simulate biomolecules in water?

Biomolecules do not function in isolation: **their native environment is aqueous.**

- Water plays an essential role in the structure, stability and flexibility of biomolecules
- It forms hydrogen-bond networks with biomolecules
- It screens electrostatic interactions
- It contributes to hydrophobic effects
- It can participate directly in binding, catalysis and conformational changes

What can we learn from hydrated MD simulations?

MD connects molecular structure, dynamics and hydration at atomic resolution.

Structural properties

- RMSD/RMSF
- Radius of gyration
- Secondary structure

Solvation properties

- Hydration shell (RDF)
- Hydrogen bonds
- Water residence time

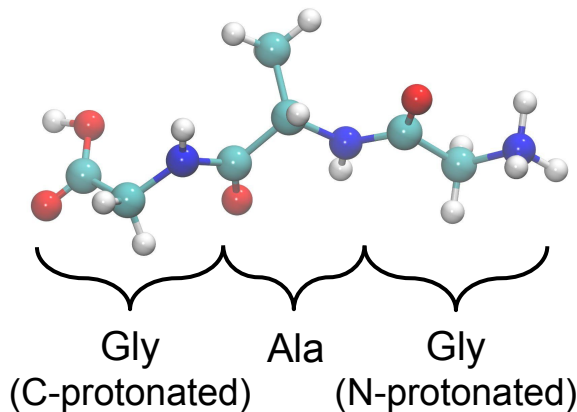
Thermodynamic and kinetic properties

- Conformational populations
- Binding affinities
- Folding rates

We will see most of these quantities in detail in the next lecture
and in practice in the afternoon exercise!

What can we learn from hydrated MD simulations?

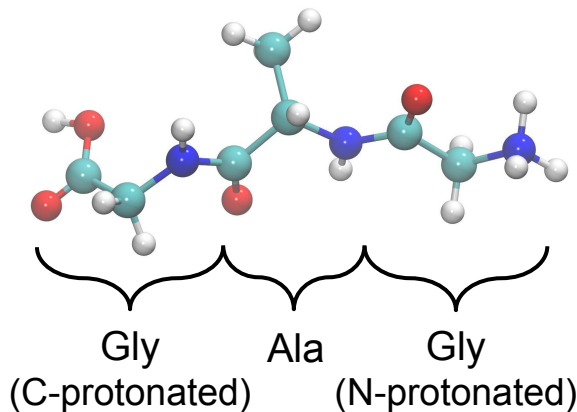
Example: MD simulations of small solvated peptides (GAG^+)



How many stable conformational states can such a small peptide have?

What can we learn from hydrated MD simulations?

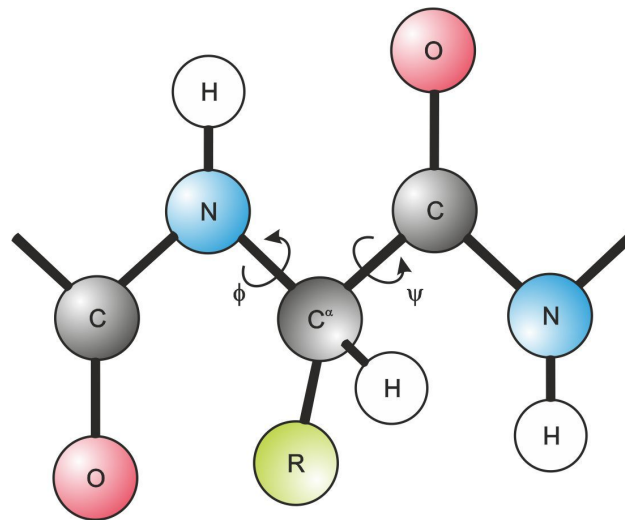
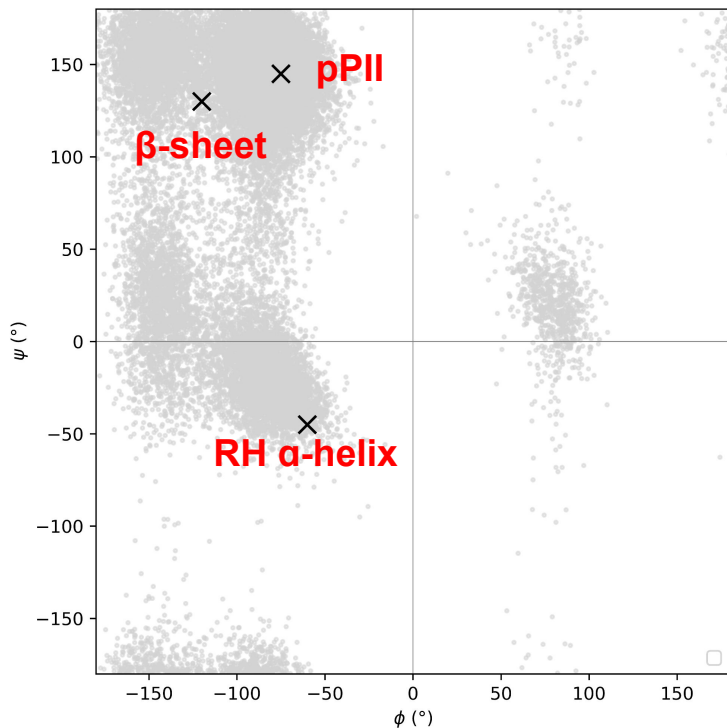
Example: MD simulations of small solvated peptides (GAG^+)



- Even a peptide this small has multiple stable conformational states
- Conformational landscape is affected by factors as hydration and temperature

GAG⁺ in aqueous environment

Two **main** populations of GAG⁺: pPll- and β -like conformations

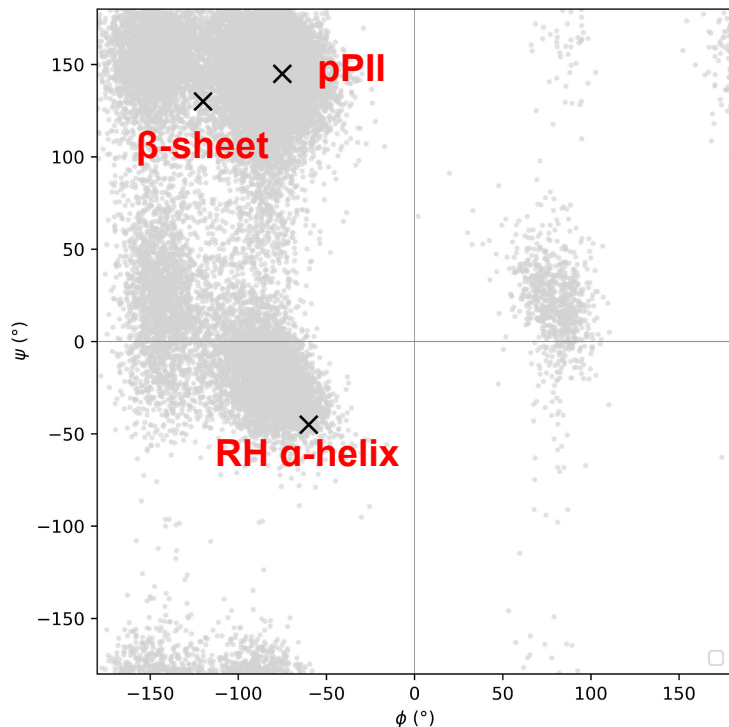


Ramachandran plot visualizes the backbone dihedral angles for amino acid residues.

backbone angles $\leftrightarrow$ secondary structure

GAG⁺ in aqueous environment

Two **main** populations of GAG⁺: pPll- and β -like conformations



pPll-like conformers are favoured at lower T,
while β -like conformers at higher T

How did we understand that?



Starting with MD
simulations!

GAG⁺ in aqueous environment

There have been a few studies on G-x-G tripeptides

The solution conformations of amino acids from molecular dynamics simulations of Gly-X-Gly peptides: comparison with NMR parameters

Author: David van der Spoel | [AUTHORS INFO & AFFILIATIONS](#)

Publication: Biochemistry and Cell Biology • May 1998 • <https://doi.org/10.1139/c98-025>

Role of Enthalpy–Entropy Compensation Interactions in Determining the Conformational Propensities of Amino Acid Residues in Unfolded Peptides.

[Siobhan E. Toal](#); [Daniel J. Verbaro](#); [Reinhard Schweitzer-Stenner](#) *

+ [Author & Article Information](#)

J. Phys. Chem. B (2014) 118 (5): 1309–1318.

<https://doi.org/10.1021/jp500181d> [Article history](#) 

A new interpretation of the structure and solvent dependence of the far UV circular dichroism spectrum of short oligopeptides[†]

[Anshuman Kumar](#) ; [Reinhard Schweitzer-Stenner](#)  ; [Bryan M. Wong](#)  

 [Check for updates](#)

+ [Author and article information](#)

Chem. Commun. (2019) 55 (40): 5701–5704.

Water-Centered Interpretation of Intrinsic pPII Propensities of Amino Acid Residues: *In Vitro*-Driven Molecular Dynamics Study

[Derya Meral](#); [Siobhan Toal](#) #; [Reinhard Schweitzer-Stenner](#); [Brigita Urbanc](#) *

+ [Author & Article Information](#)

J. Phys. Chem. B (2015) 119 (42): 13237–13251.

<https://doi.org/10.1021/acs.jpcb.5b06281> [Article history](#) 

and many more that can't fit the slide!

GAG⁺ in aqueous environment

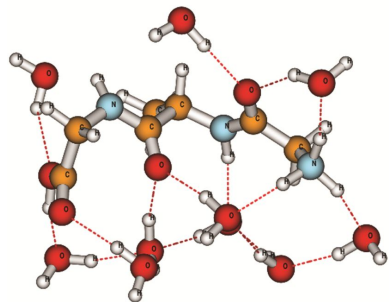
Why study the conformational landscape of these peptides?

- Small GxG peptides allow us to probe the **intrinsic conformational preference of a residue** with minimal influence from tertiary structure.
- These intrinsic preferences help us understand the **starting ensemble for protein folding** and the behavior of intrinsically disordered regions.
- Their conformational populations are strongly influenced by **hydration, temperature and side-chain chemistry**.
- Experimental measurements usually provide **ensemble-averaged observables**; MD helps resolve the individual conformational states underlying those averages.
- Because extensive spectroscopic data exist, GxG peptides are also excellent **benchmarks for force fields and water models**.

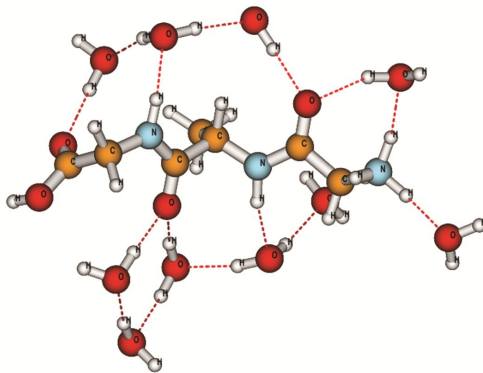
GAG⁺ in aqueous environment

Why pPII at low T and β at high T?

$$\Delta G = \Delta H - T\Delta S$$



pPII GAG



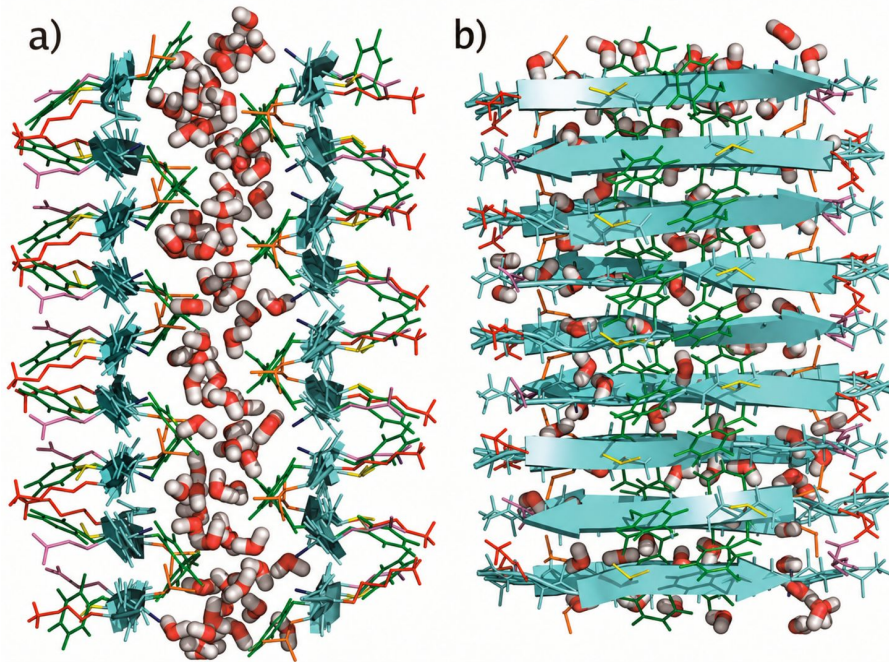
β -strand GAG

- pPII is enthalpically favored: favorable peptide–water interactions stabilize this conformation (“clathrate-like” hydration structure)
- β -strand is entropically favored: more heterogeneous hydration environment

Low T: smaller $T\Delta S \rightarrow$ enthalpic stabilization dominates $\rightarrow$ **pPII favored**

High T: larger $T\Delta S \rightarrow$ **β population increases**

From small peptides to large biomolecular assemblies



Amyloid- β 16-22 peptides

- Modern MD can treat **large solvated biomolecular assemblies** containing many peptide/protein chains and thousands of water molecules.
- We can investigate processes such as aggregation, self-assembly and hydration of interfaces.
- The system size should follow the **scientific question**: sometimes the full assembly is needed, while in other cases a smaller region or sequence is sufficient.